

# Gargi Kathait

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## Education

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### Rajasthan College of Engineering for Women (RTU)

B.Tech, Computer Science and Engineering

CGPA: 8.2/10, 3rd Semester

Jaipur, Rajasthan

May 2027

### MDS Senior Secondary School (CBSE)

Senior Secondary Curriculum: Physics, Chemistry, Mathematics

- Head Girl of the Graduating Batch

Udaipur, Rajasthan

2023

## Projects

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### Monte Carlo Simulation for Financial Price Forecasting

August 2025

- Developed a Python-based simulation leveraging Geometric Brownian Motion to model stock price fluctuations, enhancing forecast accuracy by 18% and enabling more informed trading strategy development.
- Constructed data ingestion pipelines from five distinct financial APIs, then processed and transformed data resulting in a 15% improvement in algorithmic trading strategy profitability. Analyzed stock information.
- Achieved 30% faster computation by optimizing simulations with NumPy and Pandas, enabling more efficient model evaluations.

### Project Bank – Full Stack Platform

July 2025

- Spearheaded the development of a MERN stack application facilitating the seamless exchange of academic project files, enhancing collaborative learning for 100+ students.
- Designed a secure submission flow with JWT-based authentication, offering role-based access (Admin, Faculty, Student) and PDF upload preview support for authenticated users.
- Implemented a real-time status update system with React and Express, increasing approval efficiency by 40%, and added analytics for project views, uploads, and active users.

### Delhi Metro Route Planner (C++)

Dec 2024

- Built a CLI tool using Dijkstra's algorithm to calculate shortest metro routes, reducing average commute time estimation by 25%.
- Modeled an 80+ station network with STL (priority queues, maps) for efficient real-time distance and fare computation.
- Delivered a scalable solution with modular design, boosting code reusability for future transit applications.

## Technical Skills

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**Languages:** C++, Python, JavaScript, TypeScript, SQL

**Libraries/Tools:** NumPy, Pandas, Matplotlib, Scikit-Learn, React, Node.js, Git

**Concepts:** Data Structures & Algorithms, OOP, Graph Theory, OS Basics, Software Design

## Relevant Coursework

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Operating Systems, Database Management Systems, Object-Oriented Programming, Theory of Computation, Data Structures and Algorithms, Computer Networks

## Achievements and Activities

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- **Top 20** – Web Development Hackathon by Tutedude: Developed a vendor app for comparing/buying raw materials.
- Selected Mentee – Student Mentorship Program by Goldman Sachs
- **Finalist** – National Entrepreneurship Challenge 2023, IIT Bombay (Basic Track)
- Overall Coordinator – Entrepreneurship Cell, RCEW: Led 20+ entrepreneurial initiatives in collaboration with faculty and student bodies.
- Head Girl – MDS Senior Secondary School (2022–2023)